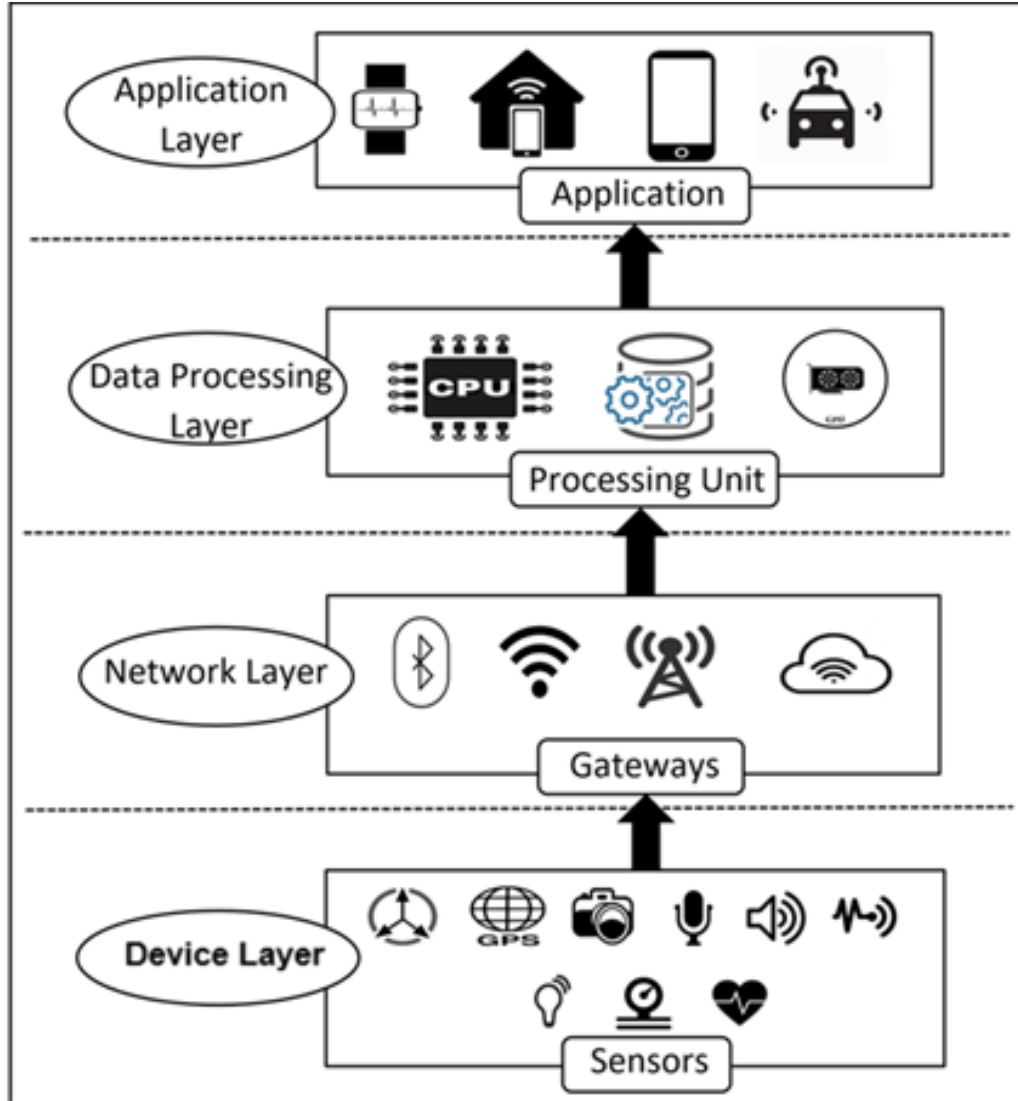
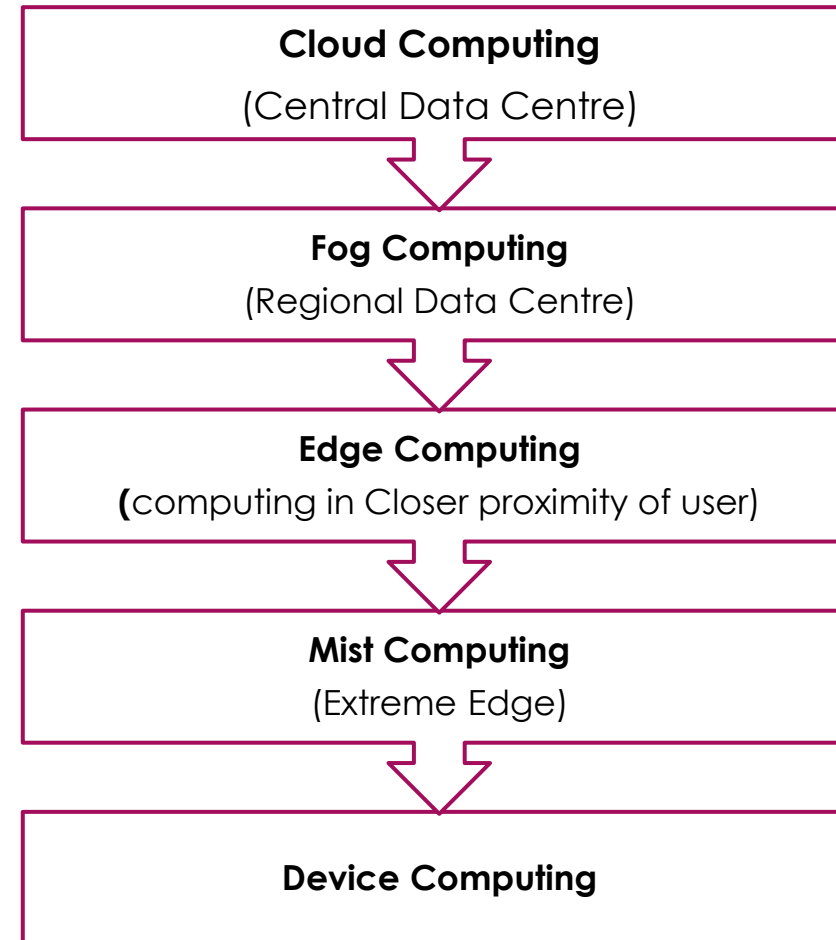


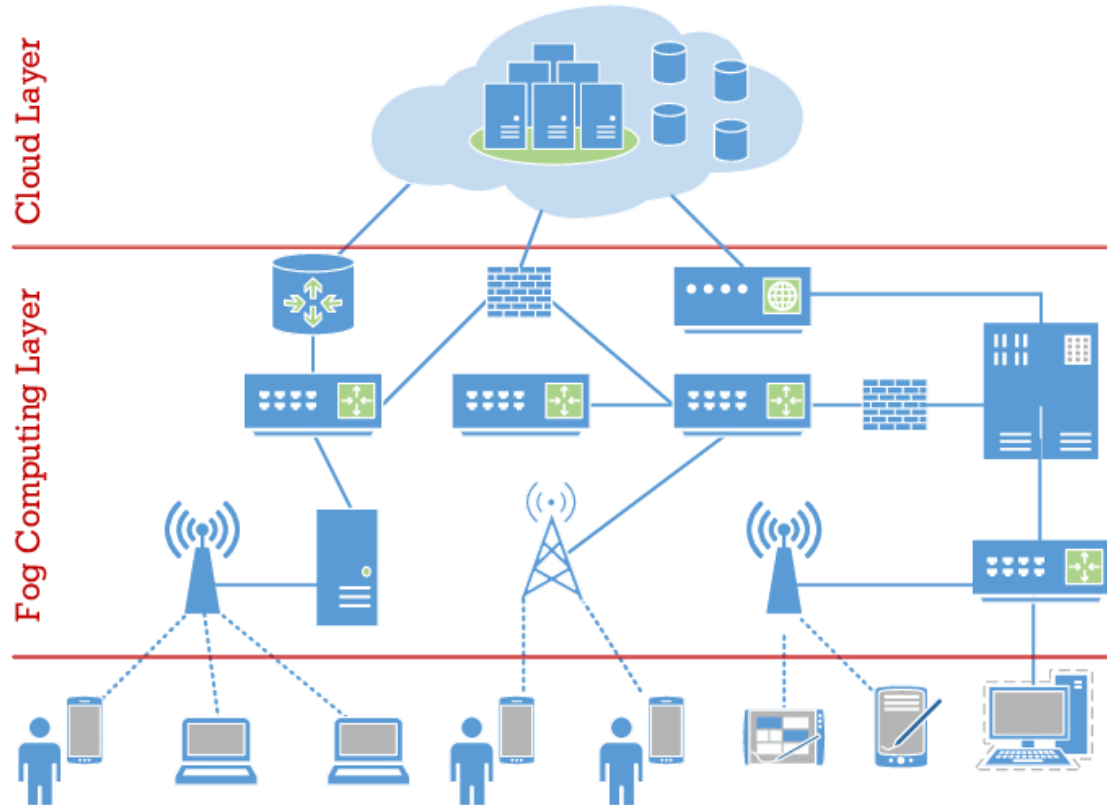
IoT – Multi Layered Architecture



Data processing happens at multiple layers



IoT – Cloud & Fog Computing



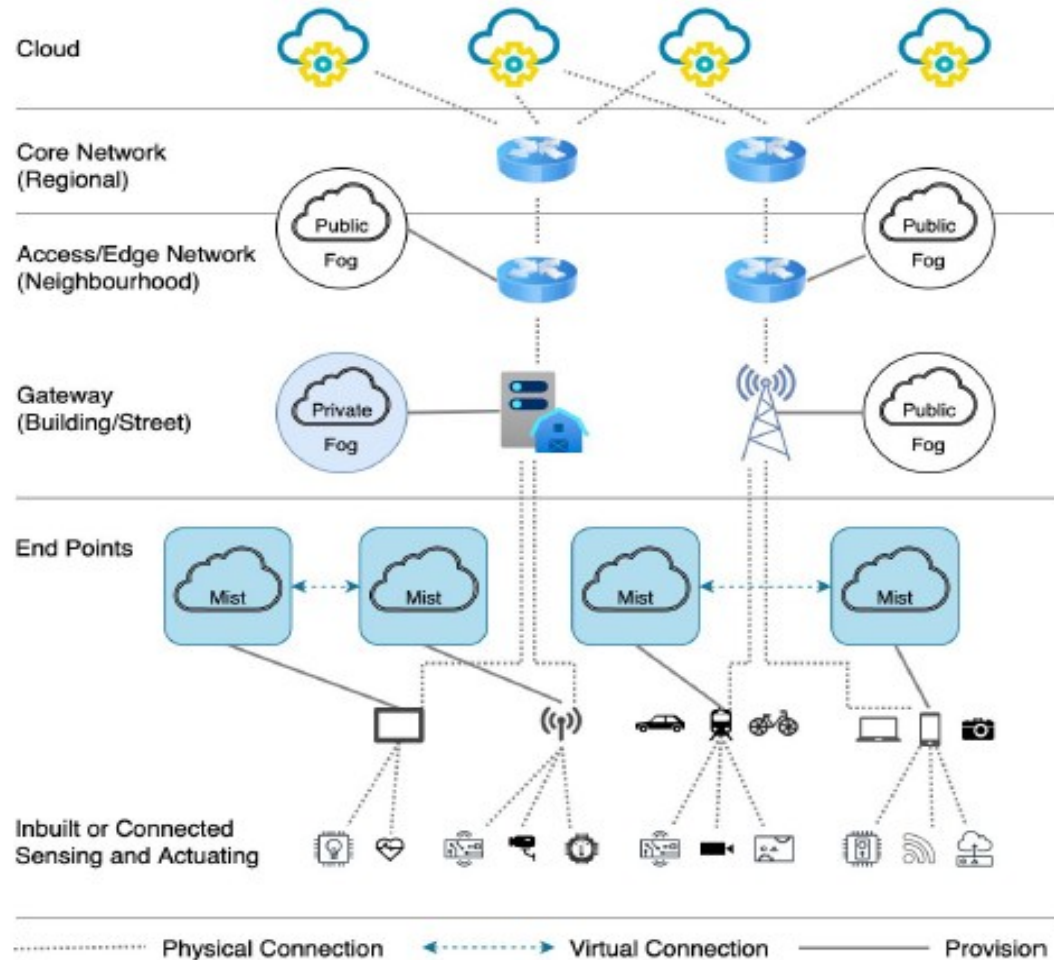
Cloud Computing

- Central Data Centre where long-term data lives
- Analyzes historical patterns, predicts congestion, generates reports for city planners, trains AI models to improve traffic flow over time
- Extremely Powerful, Scalable, and Intelligent
- But also far away, Not ideal for split-second decisions.

Fog Computing

- Regional Data Centre - Coined by Cisco for Local Network or Local Cloud
- Layer between cloud and edge to reduces load on the cloud
- Improves response time, while still handling fairly complex analytics
- IoT Friendly - Low Latency, Saves Bandwidth , Flexible storage

IoT – Edge Mist & Device Computing



Edge Computing

- Distributed computing in Closer proximity of user
- IoT Friendly –Low Latency, Less Bandwidth, pre-processes data
- Broad Definition and Implementation
- No complex / long running operations, complex analytics

Mist Computing

- Extreme Edge - computation happening almost inside the network itself
- Handle ultra-simple, ultra-fast logic like filtering noisy data or triggering an immediate alert

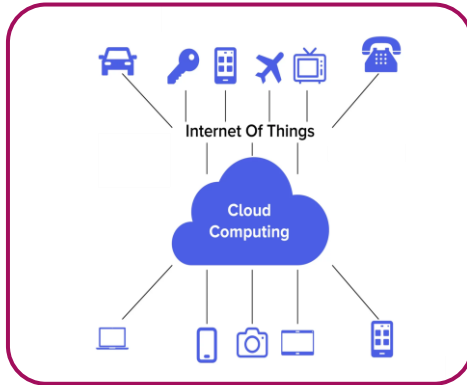
Device Computing

- intelligence built directly into the device itself

IoT – Why So many layers ?

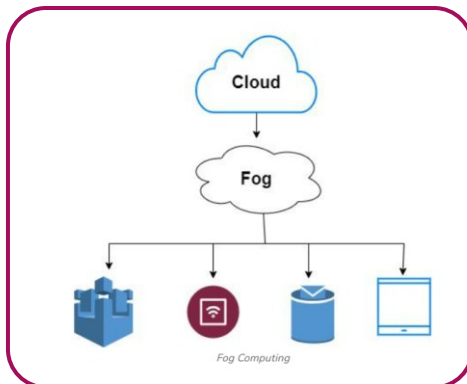
Why IoT Needed Cloud, Fog, Mist & Edge — The Evolution of Intelligence

The Cloud Era (2008–2014)



- Devices were simple with weak processor. Sensors collected data and pushed to the cloud.
- Use cases were non-critical (weather stations, fitness trackers, home automation)
- Cloud was the first brain of IoT—but it lived too far from the action

Rise of Fog Computing (2014–2018)

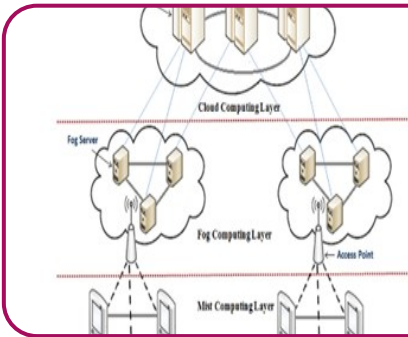


- As IoT expanded into critical domains—autonomous vehicles, industrial automation, healthcare—this cloud-only model began to crack with latency issues, bandwidth costs and slow decision making.
- This pressure led to the rise of fog computing. Fog brought intelligence closer to the ground.
- Instead of sending everything to the cloud, fog nodes performed mid-level analytics right where clusters of devices lived.

IoT – Why So many layers ?

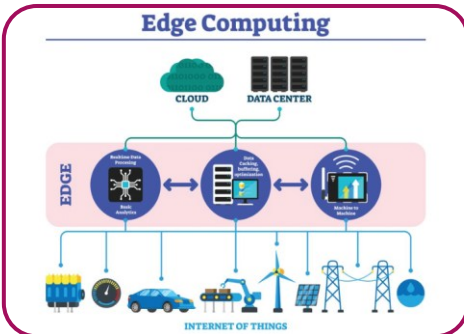
Why IoT Needed Cloud, Fog, Mist & Edge — The Evolution of Intelligence

IoT – Why So many layers ?



Mist Computing (2016–2020)

- IoT kept evolving. Sensors became smarter. Microcontrollers became more capable. And soon, even fog felt too far away.
- Tiniest layer of intelligence, living inside the sensor itself. Mist handles the first line of decision-making: filtering noise, detecting thresholds, running tiny machine learning models, and triggering immediate actions



Edge Computing (2018–Present)

- With powerful chips like NVIDIA Jetson, Google Coral, and ARM NPUs, devices could suddenly run AI models locally.
- Edge computing became the reflex of the IoT ecosystem—fast, local, and decisive. It allowed robots, drones, vehicles, and industrial machines to operate with real-time intelligence.

IoT – Why So many layers ?

Why IoT Needed Cloud, Fog, Mist & Edge — The Evolution of Intelligence



Layer	Latency	Compute	Scope	Typical Use
Cloud	Seconds+	Very high	Global	Big data, ML training
Fog	Sub-seconds	High	Local cluster	Coordination, aggregation
Edge	Milliseconds	Medium	Device / gateway	Real-time decisions
Mist	Microseconds	Very low	Single sensor	Filtering, actuation

Reason for multiple layers

Speeds

Costs

Power budgets

Privacy levels

Reliability guarantees

IoT – Multi Layered Architecture

Why IoT Needed Cloud, Fog, Mist & Edge — The Evolution of Intelligence

